WIND ENERGY
Wind Energ. 2010; **13**:573–586
Published online 28 June 2010 in Wiley Online Library (wileyonlinelibrary.com). DOI: 10.1002/we.408

RESEARCH ARTICLE

Evaluation of wind farm efficiency and wind turbine wakes at the Nysted offshore wind farm

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ABSTRACT

Here, we quantify relationships between wind farm efficiency and wind speed, direction, turbulence and atmospheric stability using power output from the large offshore wind farm at Nysted in Denmark. Wake losses are, as expected, most strongly related to wind speed variations through the turbine thrust coefficient; with direction, atmospheric stability and turbulence as important second order effects. While the wind farm efficiency is highly dependent on the distribution of wind speeds and wind direction, it is shown that the impact of turbine spacing on wake losses and turbine efficiency can be quantified, albeit with relatively large uncertainty due to stochastic effects in the data. There is evidence of the 'deep array effect' in that wake losses in the centre of the wind farm are under-estimated by the wind farm model WAsP, although overall efficiency of the wind farm is well predicted due to compensating edge effects. Copyright © 2010 John Wiley & Sons, Ltd.

KEYWORDS
wind farms; wakes; observations

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Received 17 April 2009; Revised 3 October 2009; Accepted 13 April 2010

1. INTRODUCTION

When wind turbines are arranged in large wind farms, there will inevitably be a loss of power output due to wind turbine wakes. This means that within the wind farm, wind speeds do not recover to their freestream value after encountering the first turbine (or row of turbines), and thus are lower than the freestream at subsequent turbines because kinetic energy has been extracted. Although this is a well-known phenomenon, power losses due to wakes are difficult to predict accurately due to the temporal and spatial variability of wind speed, direction, turbulence and atmospheric stability. Accurate quantification of power losses due to wind turbine wakes in differing wind climates and wind farm layouts is essential to optimal wind farm design. In small and medium size offshore wind farms (less than three rows), wake effects have been quantified experimentally,[1,2] Using the observational data, performance of both state-of-the-art wake and wind-farm models has been demonstrated to be at least satisfactory.[3,4] However, preliminary model results from the first large offshore wind farms suggests that current wind farm models using standard parameter values under-estimate power losses in large wind farms due to wind turbine wakes, leading to over-prediction of power output.[5,6] Better prediction can be obtained by adjusting parameters or using 'added roughness' over the area of the wind farm.[7,8] The implied 'deep array effect' has prompted renewed research into wake behaviour. The focus is to understand and predict power losses from wakes in large wind farms with reduced uncertainty.

In the simplest form, the velocity deficit of an individual wind turbine wake is dictated by the thrust coefficient (c_t) (and thereby to wind speed; see Figure 1) and the downstream distance, x [9]. The formula used in WAsP which has a flat velocity profile in the wake is:

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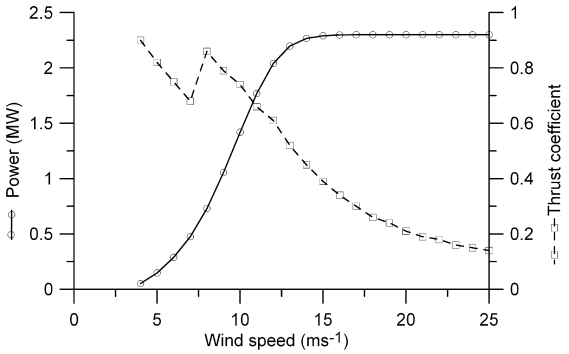


Figure 1. Power curve and thrust coefficients for the Bonus 2.3 MW turbine. Note: this uses a two-speed generator resulting in the discontinuity in c_t at approximately 6 ms^{-1}.

$$\frac{V}{V_0} = \left(1 - \sqrt{1 - c_t}\right)\left(\frac{D_0}{D_0 + kx}\right)^2 \tag{1}$$

where v is the velocity in the wake at distance x; V_0 is the freestream wind speed; D_0 is the rotor diameter; and k is the wake decay coefficient, which is set to 0.04 for offshore.

The parameterization of power output in multiple wake situations in large wind farms is an ongoing issue.[10] As has been shown, for the variation of turbulence intensity in large wind farms,[11] after c_t (see equation (1) and Figure 1), the next most important determinant is likely to be turbine spacing. A further set of variables, which are interlinked and likely of secondary importance, include turbulence and atmospheric stability. In ambient offshore conditions, turbulence intensity is also strongly determined by wind speed.[12]

Renewed interest in the measurement and modelling of wind turbine wakes—particularly in an offshore context—derives from multiple sources:

- The growth of wind farms in terms of the number of turbines deployed. Initially, wind farms were small and power losses due to wakes were relatively modest.[13] For the size of offshore wind farm currently being installed with 100 MW or greater capacity, the investment is hundreds of millions of euros, expected power output is at least 600 GWh/year, prices are at least €8.5 c/kWh;[14] therefore, even very small increases in power output that could be obtained from optimizing layouts would be significant in economic terms over 1 year.
- Wind farms were exclusively located onshore until the development of Vindeby in 1991.[15] The move offshore means wind farms are now being located in regions that experience lower turbulence[12] and where wake losses may be higher due to the influence of non-neutral atmospheric stability conditions[16] on wake propagation and recovery.
- Numerical models originally had to be linearized in order to facilitate their operation taking into account a number of other factors such as the siting of wind farms in complex terrain[17] but enhancement of computing resources means this may no longer be necessary. Explicit treatment of individual wakes and wake interactions is now possible even for large wind farms.[18–20]

Further investigation is required to quantify and understand the physical parameters that dictate the behaviour of wakes in large wind farms and to identify areas where model parameterizations may be improved. As wind farms become larger, their interaction with the boundary-layer may need to be taken into account, or the combination of wakes both downwind and, laterally, may need more sophisticated treatment in models.[21]

Here, we quantify the efficiency of the Nysted offshore wind farm based on turbine power data and compare this with standard wind farm model predictions. We begin by describing the Nysted wind farm and the meteorological context. We then use observations from the wind farm to describe wake variations in space and using conditional sampling based on different wind speed and turbulence conditions. We then present results from the application of Risø's WAsP model[22] to the wind farm and evaluate the skill of this standard approach to modelling total wake losses in large wind farms. In this paper, we report wind farm efficiency rather than wake losses because this is easier to present as a normalized quantity and is of greatest interest to wind farm developers and operators. The total freestream power is defined as the power output from the wind farm assuming all turbines produced power output without wake effects. Wind farm efficiency, e, is then defined as the sum of measured power at each turbine (p_t) divided by the total freestream power p_0:

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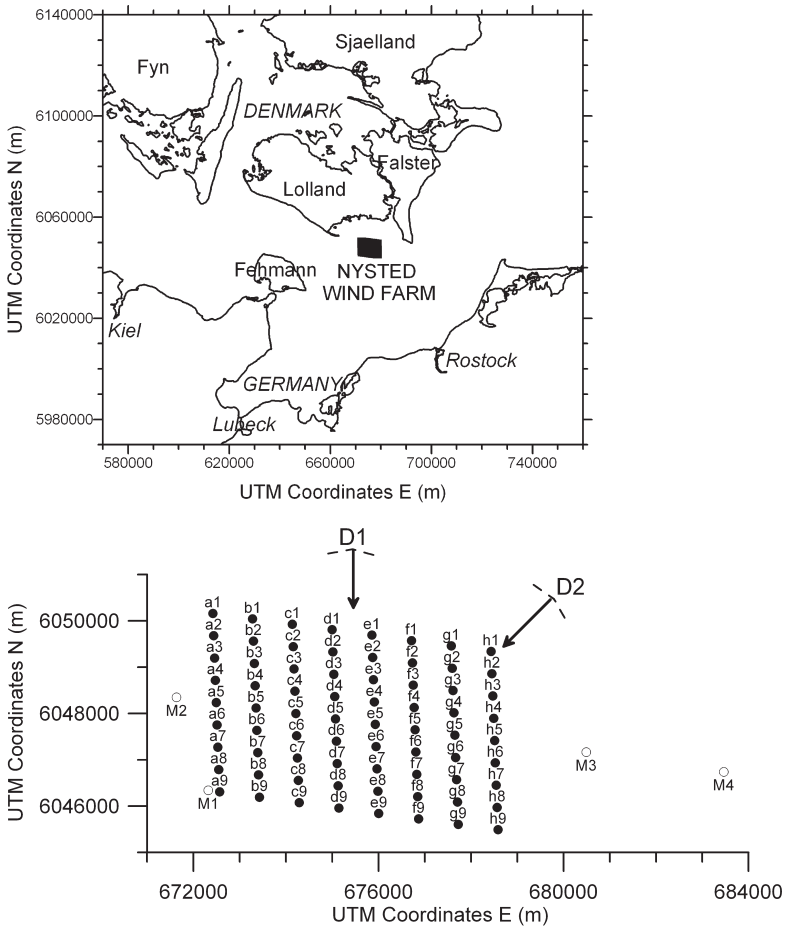


Figure 2. Above: Location of Nysted wind farm to the south of the Danish island of Lolland and to the southwest of Falster. Below: Layout of Nysted wind farm. Numbering of the turbines (filled black circles) is given (a1, a2 . . . h8, h9) while the meteorological masts (open black circles) are marked as M1–M4. Coordinates are UTM zone 32. Directions D1 and D2 are used as examples of how the freestream is derived (see section 4, point 2).

$$e = \frac{1}{n}\sum_{t=1}^{n}\frac{p_t}{p_o} \tag{2}$$

The freestream power is adjusted to account for the number of operating turbines. If all wind turbines are producing the freestream power the efficiency is 100% (and wake losses are 0%). In order to preserve data confidentiality, normalization of the results presented below has been undertaken.

2. THE NYSTED WIND FARM

Nysted wind farm was commissioned in 2004 by Energi E2 and is now owned by DONG Energy (80%) and E.ON Sweden (20%). It is the largest installed capacity offshore wind farm (165.6 MW) currently operating. It is located in a relatively enclosed sea, approximately 11 km to the south of the Danish island of Lolland, Denmark, north of the German coast (Figure 2). There are 72 turbines laid out in nine rows west to east with spacing of about 10.5 rotor diameters (D) (i.e. an inter-turbine distance of 857 m) and eight columns north to south with a spacing of 5.8 D (481 m). The turbines are Bonus 2.3 MW (see power curve in Figure 1) with a hub-height of 68.8 m and a rotor diameter of 82.4 m. It is worth noting that the turbine is stall-regulated with a two-speed generator that may impact the results of these analyses. After wind farm construction, four additional 70 m masts were erected (M1–M4 in Figure 2). Two of these are close

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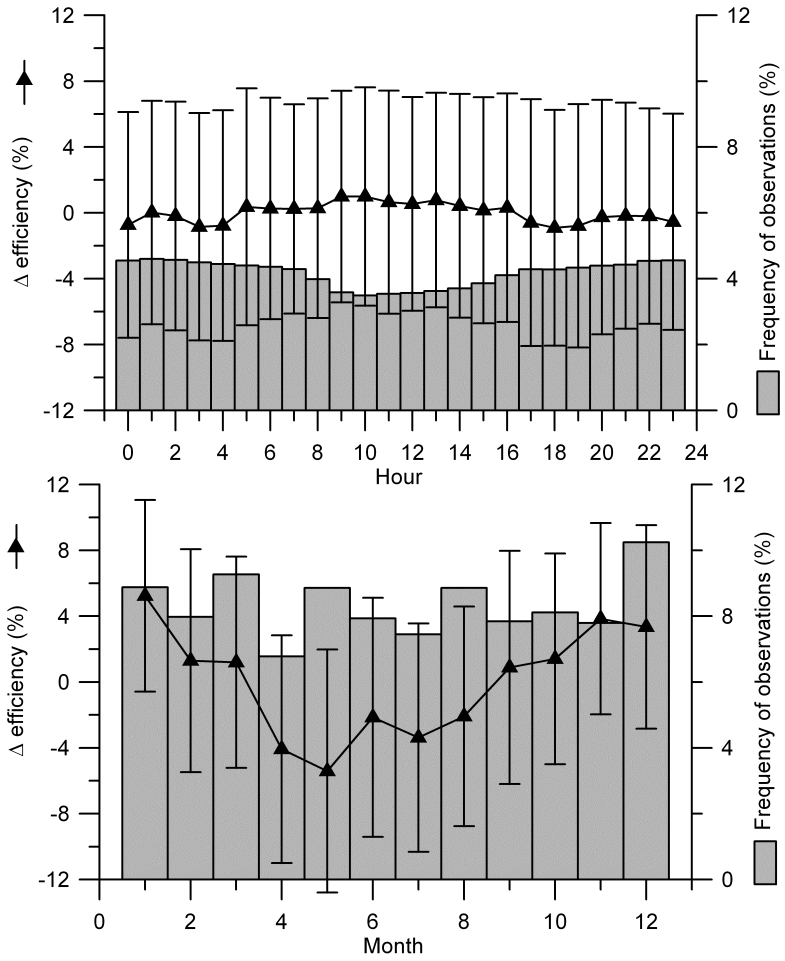


Figure 3. Difference in wind farm efficiency by hour (above) and month (below) relative to the average efficiency. Error bars indicate 0.5* standard deviation on either side of the mean. The number of total observations from each hour and month is shown by the grey bars and is presented as a percentage of the total observations.

to or upwind of the wind farm in the prevailing south-westerly wind direction (M1 and M2). The remaining masts (M3 and M4) are downwind of the array in the prevailing wind direction at distances of 2 and 6 km, respectively. For this research, 10 min average power, yaw and status signal from each turbine were analysed for the period June 2004–May 2006. Meteorological data were utilized from all four post-construction masts where wind speed profiles, direction and temperature measurements were also obtained from the SCADA database, ensuring that all data are time synchronized.

Two years of 10-min average power output from each wind turbine in the array are used here as the primary data source. These data and the ancillary meteorological observations were subject to the following conditional sampling:

- The meteorological data were used to compute atmospheric stability and ambient turbulence. Data are selected from the mast, which is determined by direction to be in the freestream (either M1 or M3; see Figure 2) and from the top height (69 m), which is also used as the freestream wind speed. Ambient turbulence is calculated as the standard deviation of wind speed in each 10 min period divided by the mean wind speed in that period. Stability is calculated using the Monin-Obukhov length[23] using the approach described in Barthelmie *et al.*[16] The 10 min period is discarded if the iteration to compute the stability parameter fails to converge (mainly at low wind speeds).
- Data from individual turbines were excluded from the analysis if the turbine did not produce power and if the status signal indicates an issue at the turbine.
- Data from an entire 10 min period were excluded if the freestream wind speed at hub-height is less than 5 ms^{-1}, or fewer than half the wind turbines in the wind farm were operating with a normal status code.

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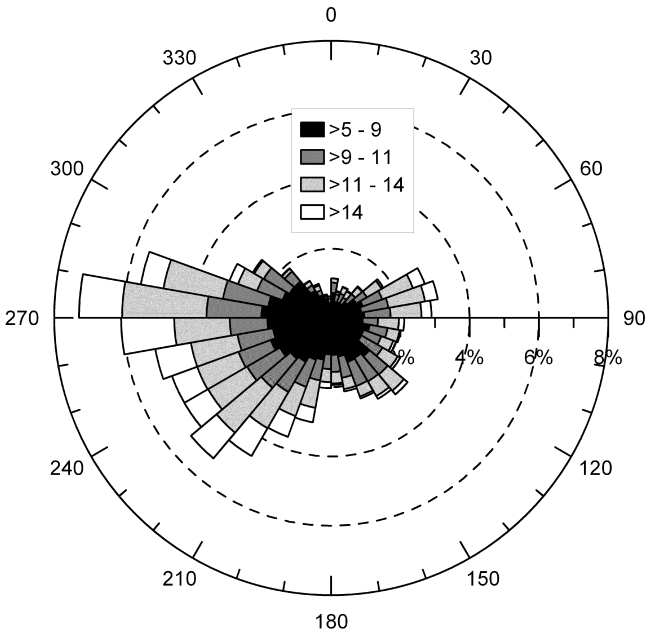


Figure 4. Directional distribution of wind speeds above 5 ms^{-1} at the Nysted wind farm. Data selection criteria are as described in section 3. Shading shows wind speed class (in ms^{-1}).

The climatological representativeness of the remaining observations was checked by plotting frequency of valid observations by hour of the day and by month (Figure 3). Recall that as described above, wind speeds below 5 ms^{-1} are not included in this analysis, thus the cumulative frequency of wind speeds sums to 100%, but the frequency shown by hour or month reflects only that fraction of 10 min periods where the wind speed equalled or exceeded 5 ms^{-1}. As shown, observations are fairly equally distributed by with at least 2500 observations for each hour of the day. As expected, summer months are slightly under-represented in the data base of 'valid' observations because the mean wind speed is lower, and hence more observations fail to meet the third criteria. Nevertheless, there are at least 4900 observations for each month of the year. Directional distribution in wind speeds above 5 ms^{-1} is shown in Figure 4 and demonstrates the importance of westerly and south-westerly flow in the power producing wind speed classes in common with most sites in northern Europe. Wind speeds in this sector also have long sea fetch as shown in Figure 2.

3. CALCULATING THE FREESTREAM POWER OUTPUT

Before wake losses can be calculated, the basic question 'What is the freestream wind speed or power output?' must be answered, since the power output computed or observed under non-wake conditions is the reference used to compute the wind farm efficiency. Under the assumption that no gradients in wind speed are observed across the wind farm, if no wake conditions exist and the freestream wind speed is thus observed at each turbine, then the efficiency will be 100%.

There are three reasonable possibilities for calculating the freestream wind speed and, hence, power output:

- Use the observed wind speed at turbine hub-height as measured at the nearest meteorological mast, which is not in the wake of any wind turbine. The main advantage of this approach is that the meteorological mast measurements are independent of the power output measurements from the turbines. However, the meteorological mast may be located some distance from the wind farm (as would be the case for easterly flow at Nysted when M3 would be the freestream mast). Further, to calculate the freestream power output the standard power curve must be applied to that wind speed (see Figure 1). As discussed in Barthelmie *et al.*,[1] there can be substantial differences between the actual power output for a given wind speed and that predicted using the power curve. We assume here that if there are differences, these are either uniform across the wind farm (e.g. due to air density) or random.
- Use the average power output in the row of wind turbines that is perpendicular into the prevailing wind direction and hence experiences undisturbed flow. The main advantage of this approach is that it is a direct measure of the power output from the wind turbines. However, as discussed by Jensen,[24] the highest power output is frequently

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observed in the corners of the array, so assigning the turbines to be used in the freestream calculations dictates the computed efficiency. This approach is applied here using eight directional sectors. If the wind direction (including directions ±22.5°) was perpendicular to the row or column, the average power from all operating turbines in the row or column was used. For example, as shown in Figure 2 as direction D1, if the flow is from the north, power from turbines in row 1 (a1..h1) are utilized. For directions perpendicular to the array corners (±22.5°), the corner turbine plus four turbines in the adjacent row and the column were used. For example, as shown in Figure 2 as direction D2, for flow from the north-east, power from turbines (e1..h1 and h2..h4). In both cases, at least four turbines had to be operating to calculate the freestream power.

- Use the highest power output measured at any turbine in the array. The main advantage of this approach is that it is easy to calculate, and ensures the resolved wind farm efficiency is below 100% (the theoretical limit). However, the highest power output may not necessarily be in the freestream depending on a number of factors such as wind speed gradients across the wind farm (due, for example, to the passage of fronts or other mesoscale phenomena) or the availability (operation) of individual turbines.

The choice of freestream values makes a fundamental difference to the magnitude of wake losses calculated. Using either of the first two methods, the measured power output at individual turbines and/or the average power from the wind farm (i.e. the efficiency) is frequently above 100%. This is due to a number of factors that may include:

- spatial gradients in wind speed over the wind farm[25]
- the pattern of non-operating turbines
- at lower wind speeds some wind turbines may be operating at different generator speeds [the wind turbine is active stall control with two-speed generator (Figure 1)]
- in the first method, differences between the manufacturer's power curve and turbine behaviour onsite.

If the individual turbine power output is above the freestream power, the overall effect when averaging is to decrease apparent power losses due to wakes. In contrast, using the third method, the measured power output is guaranteed to be lower than 100%, but the disadvantage is that the magnitude of wake losses is exaggerated because the freestream power is set to the highest power observed, which may be due to spatial variability of wind speeds over the wind farm rather than to the turbine being located in the freestream flow.

Wind farm efficiency calculated using the three methods is quite different; the first method gives an efficiency that is 4% lower than the second method and 12% higher than the third. As mentioned above, the first two methods give higher efficiency because wake losses are offset by efficiencies greater than 100%. On the other hand, the third method assumes all turbines should always produce the same as the freestream power, so these include other losses than wakes, e.g. turbines running less efficiently for other reasons (although the status signal was checked in pre-processing). In the subsequent analysis, the second method is always used to calculate the freestream values, but efficiencies higher than 133% were discarded. These amounted to 210 cases, leaving more than 72 000 observations for the calculations.

4. OBSERVED EFFICIENCY OF THE WIND FARM AND PHYSICAL DEPENDENCIES

4.1. Diurnal and seasonal variations

Average efficiency by hour of the day and month of the year is shown in Figure 3. In this figure, as in all presented here, wind farm efficiency is given as a difference from average efficiency. As expected, wind farm efficiency does not vary a great deal by hour of the day but is higher in the highest efficiency winter months (December and January) than in the lowest efficiency summer month (July). As discussed in section 1, the strongest determinant of wake losses is the turbine thrust coefficient c_t. For the Bonus 2.3 MW turbine, the thrust coefficient is highest at wind speeds up to 10 ms^{-1} (as shown in Figure 1), and then decreases as wind speeds increase. Thus, it is to be expected that average wake losses are higher in the summer when wind speeds are generally lower than during winter. Accordingly, average wake losses in May are about 10% higher than they are in January.

4.2. Spatial variability

The distribution of wake losses through the wind farm is difficult to predict using existing models and standard parameters. To the first order, it depends principally on the distribution of wind speeds by wind direction and the spacing of the wind turbines in different directions. The wind farm was designed to minimize wake losses by having larger spacing

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in the west–east direction (10.5 D) (along the axis of the prevailing flow, Figures 2 and 4) than in the north–south direction (5.8 D).

Figure 5 shows the calculated mean efficiency of each turbine in the wind farm. Wind speeds at Nysted are influenced by the nearby land, particularly in the north and north-east sectors (Figure 2). This also influences the quantification of power losses due to wakes since some of the spatial variability in power production derives from the presence of gradients in wind speed over the wind farm impacts due to coastal effects.[25] As shown, the highest turbine efficiencies are present in the outermost rows, particularly those to the south and west that experience higher wind speeds and do not experience wake losses in the prevailing wind direction (Figure 4). Although efficiency is not a direct measure of wakes because it also includes the impact of wind speed and direction and spatial gradients, it is apparent that there are substantial differences in the average wake losses experienced in the centre of the wind farm and those which, at least part of the time, experience freestream winds. This is further examined in section 6.

Figure 6 shows average efficiency for the whole wind farm by wind direction. Highest efficiencies are found for the south-westerly directions which have the highest wind speeds (Figure 4) and relatively large turbine spacing, while both southerly and northerly directions, where the turbine spacing is 5.8 D, have relatively low efficiencies (see 180/190° and 360°). Note also the lack of symmetry around the 'direct wake' directions 0° and 180°, which is in part due to the wind speed and direction distribution impact on efficiency. Efficiencies by wind direction were selected by wind speed ranges. This confirms that wind speed (and thus, c_t) is the primary determinant of wake losses. Above 15 ms^{-1}, wind farm efficiencies are above 95% in all directions. Below 10 ms^{-1}, wind farm efficiencies are lower in almost all directions than efficiencies for wind speeds in the 10–15 ms^{-1} range, regardless of wind direction and thus turbine spacing.

4.3. Directional distribution of efficiency and relationship to turbine spacing

The issue of optimal turbine spacing relates to both wake-induced power losses and increased loads.[11] These effects have proved difficult to quantify, due in part to the lack of wind farm data but also due to the relatively small spacing signal compared to the variability in the freestream due to wind speed, direction, turbulence and other factors. This analysis uses data from Nysted to evaluate whether it is possible to quantify the difference in wake losses as a function of turbine spacing.

As a first approach to this question, the minimum distance between each turbine in the array was computed in 10° directional sectors. It should be acknowledged that this analysis does not account for the width of wakes or their expansion rates (and hence only accounts for direct wake impacts), and thus turbine separations beyond 15 D are discarded. Ignoring spacing greater than 15 D where the uncertainty is large, a linear fit can be made and the normalized efficiency extrapolated back to smaller turbine separations. Data are selected according to the criteria given in section 3. Figure 7 shows the efficiency (difference from the average) related to the minimum turbine separation (calculated as a function of wind direction sector). For wind speeds from 5–15 ms^{-1}, the variability around the best-fit line (computed using least squares linear regression) is relatively small and 68–76% of the variability of the efficiency is attributable to the turbine spacing. The linear fit to data for wind speeds from 5–15 ms^{-1} suggests an increase in normalized wind farm efficiency of approximately 1.3% for each increase in turbine spacing of 1 D rotor diameter. For wind speeds above 15 ms^{-1}, there is no relationship between spacing and normalized efficiency in accord with the small influence of wakes at high wind speeds.

The results shown in Figure 7 are the first direct empirical quantification of the role of turbine spacing on the power output in an operational large offshore wind farm. However, it is important to recall that they are specific to the Nysted wind farm. The results are not only dependent on the specifics of the wind farm (turbine type, operation and so forth), but also the prevailing meteorology (as noted above, the efficiency is strongly dependent on wind speed and, as shown below, is also a function of prevailing atmospheric stability and ambient turbulence). Further analyses of other wind farms are necessary to test the generalizability of these results.

4.4. Influence of atmospheric stability and turbulence intensity

As is shown above, it is difficult to separate meteorological and physical influences on wind farm efficiencies/wake losses. A further set of relationships that needs to be explored is between wind speeds, ambient turbulence and stability.[26] Ambient turbulence intensity and atmospheric stability are inter-related. It might be expected that wind farm efficiencies are higher in high turbulence conditions because these should be associated with enhanced mixing in the surface boundary-layer, which will transfer momentum downwards from higher wind speeds aloft. Accordingly, in stable conditions when turbulent mixing is inhibited and the near-surface layer has reduced exchange with layers above, wind farm efficiencies might be expected to be lower. Conversely, in unstable conditions, efficiencies might be expected to be higher. Complicating these relationships offshore is the relationship of both turbulence and stability with wind speed. Turbulence intensity

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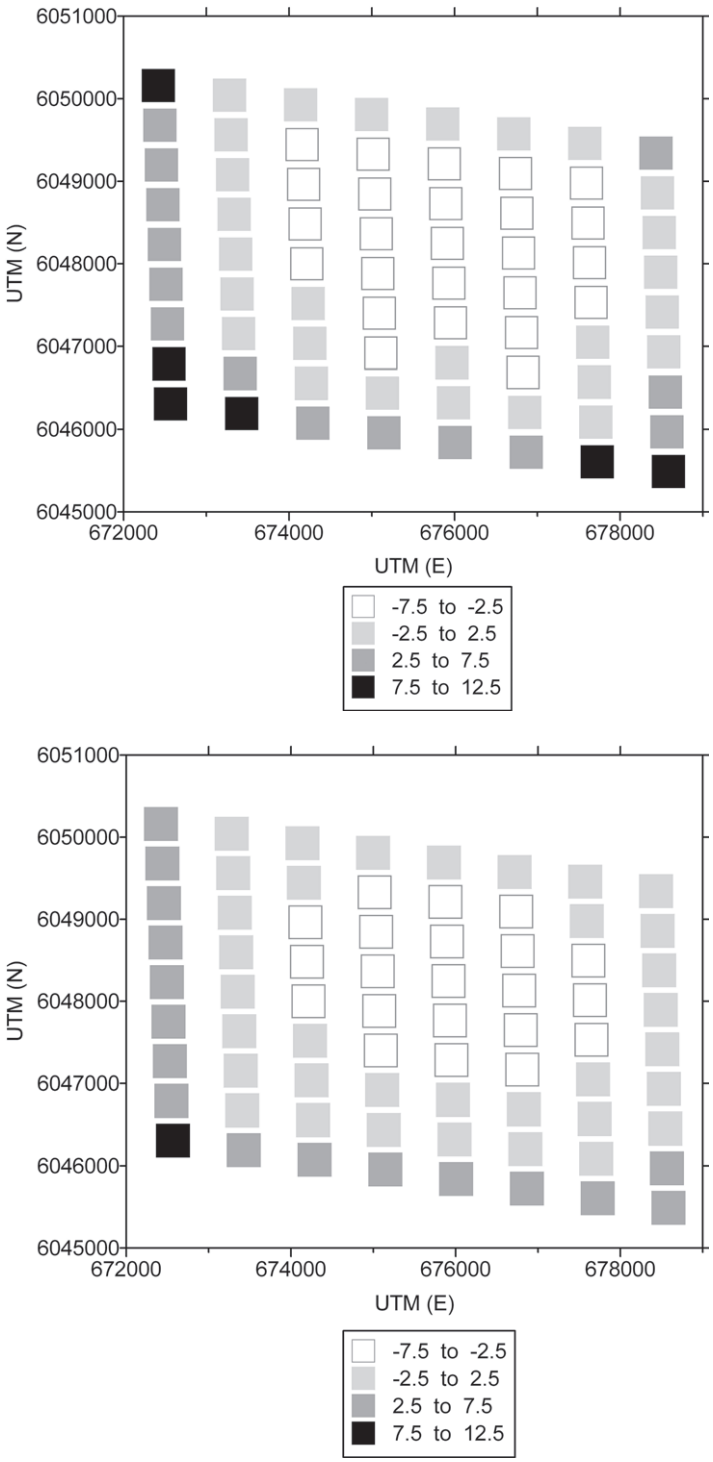


Figure 5. Average difference in turbine efficiency (%) from the array average. Above: Observations. Below: Predictions using the WAsP model with wake decay coefficient k = 0.03.

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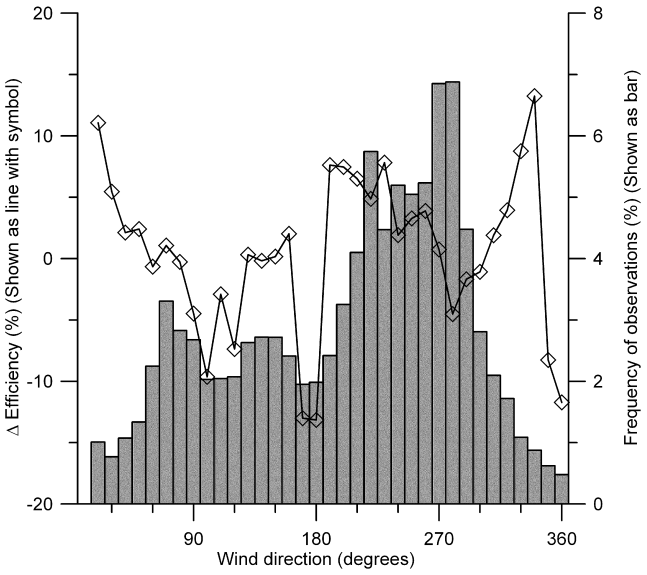


Figure 6. Change in wind farm efficiency by freestream wind direction. Wind farm efficiency (shown as the line with symbols) are shown as the difference from the array average. The relative frequency with which each wind direction was observed is shown by the vertical bars.

offshore is relatively high at low wind speeds (due to thermal generation of vertical mixing), decreases to a minimum around 8–12 ms^{-1} and then increases in response to mechanical (wave) generation.[27] Additionally, it is difficult to sustain either stable or unstable conditions offshore as wind speeds increase.[26] Above about 15 ms^{-1}, near-neutral conditions dominate. In the following, we examine the relationships between wind farm efficiency, turbulence and stability.

Table I shows normalized wind farm efficiencies by stability class for moderate wind speed classes (i.e. freestream wind speeds of 5 to 10 ms^{-1}). As shown, observations between 5 and 10 ms^{-1} can be broadly divided into 20% neutral, 40% stable and 40% unstable classes. The question arises 'Are differences in wind farm efficiency really due to stability, or to wind speeds, or to some other proxy such as turbulence intensity?' Although wind speed exerts the primary influence, differences in efficiency between the stability classes are preserved as wind speed changes. While wind farm efficiencies in unstable classes are only marginally higher than those in the near-neutral class, efficiencies in stable conditions are always lower than those in the near-neutral class consistent with the physical explanation given above.

To investigate whether ambient turbulence levels are in part responsible for some of the differences portrayed in Table I, an additional analysis was undertaken. Average ambient turbulence intensity for all observations in the analysis is 5.6% (wind speeds lower than 5 ms^{-1} are excluded). Turbulence varies according to stability class being 4.6, 6.0 and 6.4% in stable, neutral and unstable conditions, respectively. Normalized wind farm efficiencies are shown by ambient turbulence in Figure 8. As ambient turbulence increases, wake dissipation rates increase and the wind farm efficiency increases as expected. However, as with stability, this effect on wind farm efficiency is secondary to wind speed impacts.

5. MODELLED WIND FARM EFFICIENCY

Given the magnitude of power losses due to wakes in large wind farms, there is considerable interest in evaluating the predictive capability of wind farm models. A number of models are being evaluated against case studies from large offshore wind farms at both Horns Rev and Nysted,[28] results of which indicate a wide range of model skill. As a preliminary evaluation of model ability to capture wake behaviour, we focus here on the WAsP model,[22] which is widely used by the wind energy industry.

The wake model in WAsP has a straightforward parameterization of wake expansion, in which the velocity deficit at a given distance is dependent on the thrust coefficient, turbine rotor diameter and the wake decay coefficient (equation 1). The wake decay coefficient (k) describes the rate of wake expansion and is thus an abstraction of roughness and stability influences on turbulence and hence wake dissipation. Recommended values for k are 0.075 for land and about 0.04–0.05 for offshore.[22] The wake decay constant is related to roughness and thereby to average turbulence intensity but no account is taken of varying turbulence or atmospheric stability. The value for offshore has been found to work well for small wind farms,[4] but to under-predict wake losses (over-predict power output) at the large offshore wind farm at

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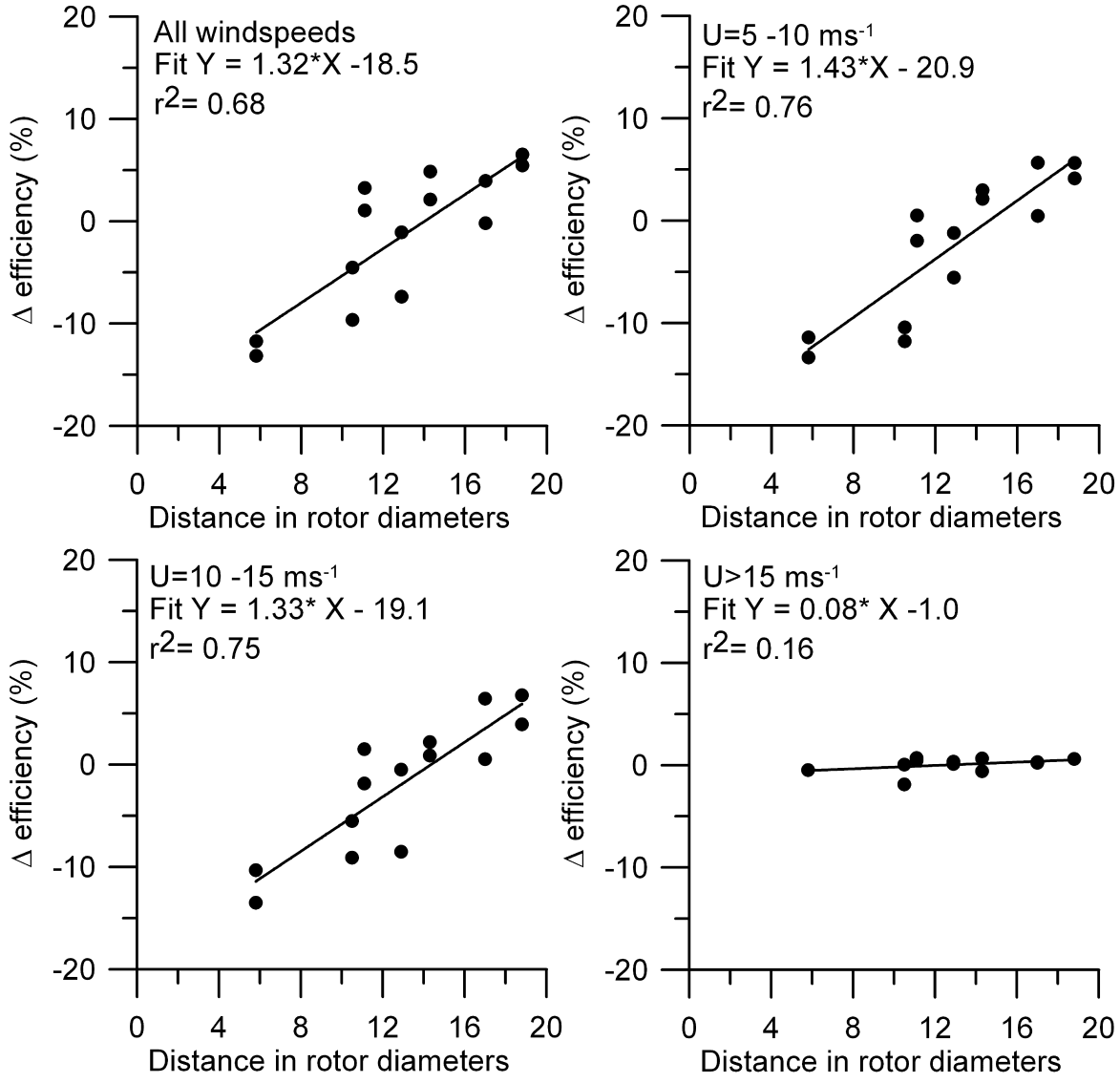


Figure 7. The difference in turbine efficiency from the array average by minimum turbine separation (expressed as the distance in rotor diameters) for different wind speed classes. Linear fits to the data are shown along with the equation for the fit and the variance explained.

Table I. Wind farm efficiencies in different wind speed and stability conditions. The top figure in each row is the difference in efficiency from the neutral value in each wind speed class (%). The figure in brackets below shows the percentage of observations in each wind speed group ascribed to different stability classes. Recall that wind speeds below 5 ms^{-1} are not included in this analysis. The definition of the stability class is shown according to the value of the Monin-Obukhov length (L).

	5–6 ms^{-1}	6–7 ms^{-1}	7–8 ms^{-1}	8–9 ms^{-1}	9–10 ms^{-1}
Stable	−6.1	−5.7	−4.7	−7.2	−8.1
0 < L < 1000 m	(34.7)	(38.7)	(37.4)	(39.0)	(38.7)
Neutral	0.0	0.0	0.0	0.0	0.0
|L| > 1000 m	(21.3)	(18.7)	(19.8)	(23.1)	(26.4)
Unstable	3.0	1.3	0.7	1.0	0.8
−1000 < L < 0 m	(44.0)	(42.6)	(42.8)	(37.9)	(34.9)

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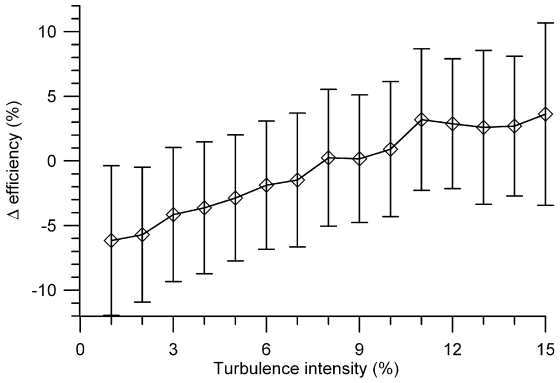


Figure 8. Difference in wind farm efficiency (relative to the average) by ambient turbulence intensity. Error bars shown are 0.5* standard deviation on either side of the mean.

Horns Rev[29] (off the west coast of Denmark). A wake decay coefficient of 0.03 was found to be a better fit to Nysted data.[6] Herein we evaluate whether WAsP is able to replicate the 'deep array' effect (i.e. minimum efficiencies in the centre of the array) evident in the observations (Figure 5). The 'deep array' effect is here understood to mean that there are additional wake losses in the centre of large wind farms which are not captured by current wake models using standard parameterizations. It has previously been speculated that this is due to lack of feedback between the turbines and the overlying boundary layer or to wake merging downwind or laterally within the wind farm.[10] For multiple wakes in WAsP, it is formulated that the total energy deficit at a wake position is equal to the sum of the individual energy deficits,[9] and the wake velocity, V, can be determined from the individual wake velocities, V_1 and V_2 and the freestream velocity V_0:

$$\left(1-\frac{V}{V_0}\right)^2=\left(1-\frac{V_1}{V_0}\right)^2+\left(1-\frac{V_2}{V_0}\right)^2 \tag{3}$$

To compare predicted and observed wake losses, WAsP simulations were performed using the data set from Nysted as described in section 3. Since the input wind speed to WAsP is taken from the freestream mast for the same observations as used in the wake analysis, the wind climate used is identical. However, in comparing models and observations, there are other effects to be considered such variation of wind speed over the wind farm due to coastal effects (discontinuities in roughness, temperature etc). WAsP has a short wind speed recovery distance from the coast such that the predicted variations in power output over the Nysted wind farm are less than ± 0.2%. This is likely to under-represent the actual wind speed gradients over the Nysted wind farm, particularly in stable conditions. Obviously, these gradients vary by fetch and, hence, wind direction, with largest gradients being associated with flow from north and for some turbines for easterly flow.[25] Another difficulty in evaluating model performance is in correcting for missing observations. The procedure adopted here is to calculate the total energy output over the whole period at each turbine by summing the power output in each observation and then divide this by the number of observations at that turbine to give an average energy output. This can then be multiplied by the number of hours in a year to get a 'total' energy output. However, this assumes that the missing data are randomly distributed with wind speed and that the absence of wake effects from non-functioning turbines only introduces stochastic effects to the analysis.

To evaluate whether WAsP accurately predicts the spatial variability of wake losses, the efficiency computed from both simulations and observations were calculated relative to average efficiency. As shown in Figure 5, WAsP predicts lowest efficiency in the centre of the wind farm due to the cumulative wake losses from each direction. This pattern is very similar to normalized efficiencies from the observations as shown in the upper panel of Figure 5. It is important to note here that this is not a validation of the absolute efficiency since both data sets are standardized to their own average, although the overall efficiency is well predicted by WAsP. While WAsP captures much of the observed spatial variability (as shown in Figure 5) the range of efficiencies is slightly larger in the observed dataset than predicted by WAsP. The observed efficiencies are very high in the corners of the array and lower in the centre, while WAsP does not predict such a wide range. This can be clearly seen in Figure 9, where the turbine efficiencies are plotted individually. This is an indication that that WAsP does not fully capture wake losses at the centre of the array (whether a wake decay coefficient of 0.03 or 0.04 is used) but compensates to some degree by applying larger wake losses/lower efficiencies at the array edges and corners. Thus, although there is some evidence for the 'deep array effect' it is relatively minor compared to the overall uncertainty in wake measurement and prediction.

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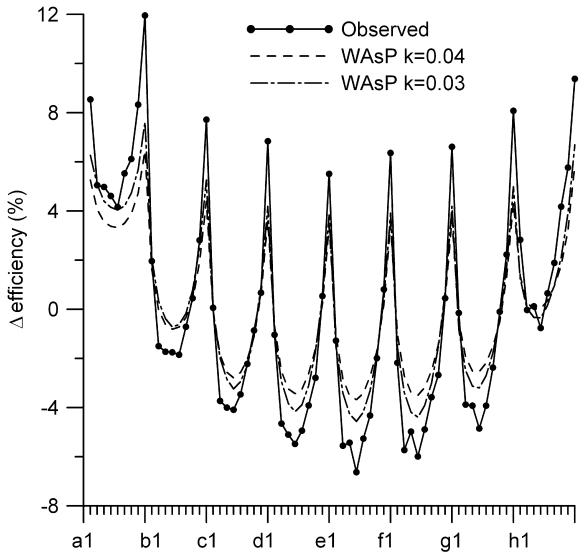


Figure 9. Difference in turbine efficiency from the array average by turbine. The x-axis gives the label for the first turbine in each column as shown in Figure 2 where each column has turbines *a1*, *a2*, *a3* . . . *a9* and then *b1*, *b2*, *b3* . . . *b9* to *h1*, *h2*, *h3* . . . *h9*. Observations are shown as the solid line with symbol. Predictions from the WAsP model with wake decay coefficient, *k*, of 0.04 or 0.03 are shown with the dashed lines.

6. CONCLUSIONS

Careful and detailed analysis of power output data from the offshore wind farm at Nysted has been undertaken to quantify the role of wind speed, wind direction, ambient turbulence, atmospheric stability and turbine spacing in determining power losses due to wind turbine wakes and overall wind farm efficiencies. Because wind farm performance is confidential, results are mainly presented as differences from average wind farm efficiency. Wind farm efficiency is a function of not only wind turbine wakes but also turbine type, changes in wind speed due to spatial gradients and the impact of wind speed and direction distribution. Nevertheless, this analysis yields the first empirically derived relationship between wind turbine spacing and wind farm efficiency.

One of the main difficulties in describing wind farm wake losses (or efficiency) is to establish an accurate freestream. As described here, we use an average of the power output in the row or column for the four main wind directions perpendicular to the rows/columns (approximately north, south, east and west) and for the four main diagonals across the wind farm (approximately northeast, southeast, southwest and northwest), we use the average power output from the nine turbines in the outer corner of the array. This helps to limit some of the difference from using the meteorological mast data and power curve, does not depend on a single turbine and reduces to some degree the impact of having higher power output in the corners of the wind farm.

Because efficiency is most strongly determined by the thrust coefficient and thereby to wind speed, it shows some variability with season, being lowest in the lower wind speed months, but relatively little change of the course of the day. Individual turbine efficiencies are strongly related to their location in the array with differences of up to 20% between those in the centre at those at the edge. Efficiency is also determined by wind direction, partly because wind speed distributions are direction dependent, but also because of the wind farm layout with different spacing in the north–south and west–east directions. The change of wind farm efficiency is estimated to be of the order 1.3% per one rotor diameter change in turbine spacing for wind speeds up to 15 ms^{-1}. This is an approximation based on a linear fit to the data and may be appropriate only for a limited range of wind speeds, turbine spacing and for this particular wind farm climate and layout. The generalizability of this result should be tested as other wind farm data sets become available. Efficiency also depends on atmospheric turbulence and on whether the atmosphere is stable or not. These effects may be stronger at offshore wind farms because mechanically generated turbulence is relatively low but are demonstrably second order at Nysted.

Finally, the observed turbine efficiencies are compared with those predicted by the WAsP model and for a wake decay coefficient of 0.03 the whole wind farm efficiency is captured to within 0.5% of the observed value. WAsP tends to under-predict wake losses in the centre of the array (the 'deep-array effect'). This under-prediction is compensated to some degree by prediction of lower than observed efficiencies at the edge of the array, which could be interpreted as a

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lack of prediction of speed-up effects or an over-prediction of wake losses at the corners and edges of the wind farm. Nonetheless, the overall spatial variability of wind turbine efficiency is captured. Further analysis of data from other large wind farms is needed, both on- and off-shore, to evaluate whether the results presented herein are representative of large wind farms in a range of environmental contexts. Given the dependence of wind farm efficiency on turbulence and atmospheric stability better physical understanding of flow within large wind farms is required to improve model parameterizations and to provide wind farm models which can be applied to accurately predict power output for large and multiple wind farms.

ACKNOWLEDGEMENTS

Data from Nysted were provided by Energi E2 (now part of DONG Energy). Funding was provided in part from the National Science Foundation CBET-0828655 and the European Commission UPWind project SES6-019945. Thanks to two anonymous reviewers for their help in clarifying the paper.

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